



Mess Management Portal

A QR-Based Digital Mess Entry & Management System for JK Lakshmipat University

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About the Project

Context

- ✓ JK LakshmiPat University (JKLU) serves hundreds of students daily across 4 meal slots.
- ✓ Current system relies on physical ID cards for entry verification.
- ✓ Manual processes create bottlenecks, fraud risks, and administrative burden.
- ✓ No digital record of meal consumption or real-time kitchen analytics exists.

Three User Personas



Students

Digital mess pass, menu view, meal skip/book with quotas, reviews & history



Mess Staff

QR scanner for real-time verification, auto duplicate detection, time-based meal logic



Administrators

Menu management, kitchen headcount, refund ledger, notices & feedback review

Challenges with the Existing System



Inefficient Entry

Physical ID cards cause long queues at peak meal hours; proxy entries go undetected.



No Real-Time Tracking

Zero digital record of meal consumption; anomalies and patterns are invisible to management.



Food Wastage

Kitchen prepares on rough estimates without advance knowledge of actual student count.



No Meal Flexibility

Students cannot skip meals in advance or track refund-eligible cancellations.



No Feedback Channel

No structured way to report food quality issues or attach photographic evidence.



Admin Overhead

Menu, headcount, refunds and notices managed manually — time-consuming and error-prone.

Key Features of the Portal



QR-Based Entry

Digital mess pass with unique QR per student. Real-time scan & verify by staff.



Auto Meal Detection

Server-clock determines meal type automatically (Breakfast/Lunch/Snacks/Dinner).



Azure AD SSO

Only @jklu.edu.in emails allowed. OAuth 2.0 via Microsoft MSAL popup flow.



Kitchen Analytics

Tomorrow's headcount visible to kitchen for precise food preparation.



Refund Ledger

Per-student cancelled meal tracking with CSV export for accountants.



Notices & Reviews

Admin broadcast notices + student food feedback with photo evidence.

Three-Tier Architecture & Tech Stack



Presentation Layer

Frontend

- ✓ React 18 + Vite
- ✓ Material UI
- ✓ MSAL.js (Azure SSO)
- ✓ React Router v6
- ✓ Axios HTTP
- ✓ QR Scanner Library



Application Layer

Backend

- ✓ Node.js + Express 5
- ✓ Mongoose ODM
- ✓ JWT Auth
- ✓ Multer (uploads)
- ✓ RESTful APIs
- ✓ CORS Middleware

Data Layer

Database

- ✓ MongoDB Atlas
- ✓ 7 Mongoose Models
- ✓ Compound Indexes
- ✓ Cloud-Hosted NoSQL
- ✓ Atomic Operations
- ✓ Schema Validation

Backend — APIs & Database

REST API Routes

POST	/api/auth/microsoft-login	Azure SSO Login
POST	/api/mess/scan	QR Verification
GET	/api/dashboard/data	Student Dashboard
POST	/api/dashboard/toggle	Toggle Meal Skip
GET	/api/dashboard/history	Meal History
POST	/api/admin/menu	Update Menu
GET	/api/admin/headcount	Kitchen Headcount
GET	/api/admin/ledger	Refund Ledger
POST	/api/reviews/submit	Submit Review

Database Models

User	students, staff, admins + roles
MealLog	scan log — who ate & when
Menu	weekly menu per meal/day
MealBooking	skip/book decisions + quota
Booking	alternative booking reference
Review	ratings, comments, photos
Notice	admin broadcast messages

Frontend — Pages & Components



AuthGate.jsx

Login Page

- ✓ Microsoft SSO popup via MSAL
- ✓ Domain validation @jkl.edu.in
- ✓ Session stored in localStorage
- ✓ Dev mode skip-login toggle



Dashboard.jsx

Student View

- ✓ Digital Mess Pass + QR Code
- ✓ Monthly skip quota display
- ✓ Today & tomorrow menus
- ✓ Review form with photo upload



Scanner.jsx

Staff Interface

- ✓ Camera QR scan via react-qr-scanner
- ✓ Auto meal type detection
- ✓ Duplicate entry rejection
- ✓ Visual success/fail feedback



AdminDashboard.jsx

Admin Panel

- ✓ Role-gated sections (admin/contractor)
- ✓ Weekly menu management
- ✓ Kitchen headcount analytics
- ✓ Refund ledger with CSV export

Planned Enhancements & Roadmap

Week 1–2

Security Hardening

JWT-based session management, HTTP-only cookies, CSRF protection, server-side route guards.

Week 2–3

Testing Suite

Unit tests with Jest/Mocha for all controllers. Integration testing for QR scan flow. React Testing Library.

Week 3–4

Deployment

Backend on Render/Railway or AWS EC2. Frontend on Vercel/Netlify. Production CORS & env config.

Week 4

Meal Pricing

Price field in Menu model. Refund ledger calculates actual amounts based on meal prices.

Week 4–5

Enhanced Signup

Collect roll number, dietary preference (Veg/Non-Veg/Eggetarian), hostel/day-scholar status.

Week 5–6

Nutritional Info

Calories, protein, carbs, fat, fibre per serving. Daily intake summary for students.

Identified Risks & Our Responses



QR Code Sharing / Fraud



Time-based token rotation + server-side cooldown preventing same QR scanning by different devices.



Network Outage at Mess



Offline fallback: scans queued locally, synced when connectivity restored; QR cached on dashboard.



Device/Browser Compatibility



Wide device testing; manual student ID entry fallback provided for staff.



localStorage XSS Vulnerability



Migrating to HTTP-only cookies with JWT + CSRF protection + Content Security Policy headers.



Race Conditions (Concurrent Scans)



MongoDB atomic findOneAndUpdate + unique compound indexes + server-side deduplication.

What We've Built

The JKLU Mess Management Portal successfully replaces a manual, error-prone cafeteria system with a robust digital platform — eliminating proxy entries, food wastage, and administrative overhead.

- ✓ Complete backend API with 10+ endpoints & 7 DB models
- ✓ Azure AD SSO with domain-restricted access
- ✓ Real-time QR scanner with auto meal detection
- ✓ Student dashboard, admin panel & history views
- ✓ Monthly quota enforcement & refund ledger

Tech Stack

React 18 + Vite

Node.js + Express 5

MongoDB Atlas

Azure AD OAuth 2.0

Material UI

MERN Stack